

AI-BFSI Internship Project Documentation

Sample Project Documentation – Subscription Waste Detector

1. Introduction

Software project documentation communicates objectives, architecture, implementation, testing, and outcomes of a system. It provides a structured record of the development process, enabling stakeholders and evaluators to understand the project’s purpose, design decisions, implemented functionality, and workflow.

The **Subscription Waste Detector** is an AI-assisted financial analysis web application developed under the **AI Specialist BFSI Program**. It helps users identify recurring subscription expenses, monitor monthly spending, receive renewal reminders, and obtain AI-powered recommendations for optimizing recurring financial commitments.

This document presents a consolidated overview of the complete project, summarizing the problem addressed, solution, architecture, implementation, testing, technologies used, and final outcomes.

2. Project Overview

Attribute	Details
Project Name	Subscription Waste Detector
Project Category	AI-Assisted Financial Analysis
Domain	Banking, Financial Services and Insurance (BFSI)
Platform	Streamlit Web Application
Programming Language	Python
AI Integration	Groq API
Development Environment	Visual Studio Code

3. Problem Overview

The rise of subscription-based services has created financial challenges for consumers. Automatic renewals often lead to unnoticed recurring payments, and traditional expense trackers provide limited visibility into subscriptions.

The Subscription Waste Detector was designed to address these challenges by improving subscription visibility and supporting informed financial decision-making.

4. Project Objectives

- Analyze uploaded expense records
- Detect recurring subscription payments

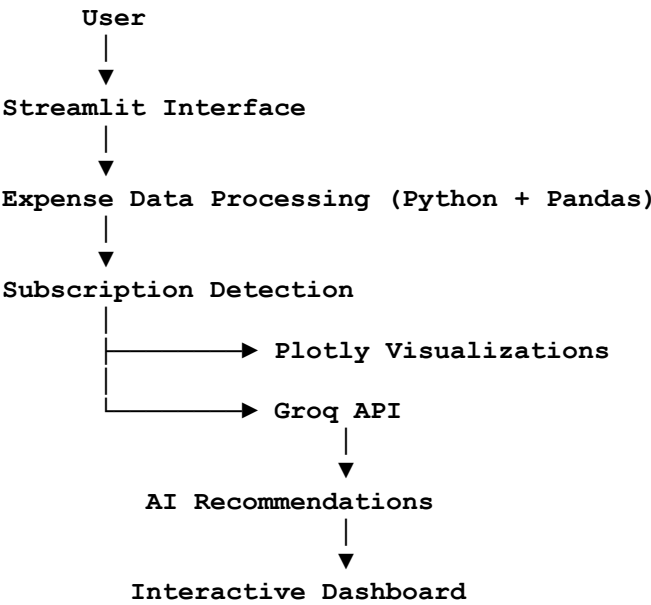
- Present subscription spending through dashboards
- Display renewal reminders
- Generate AI-powered recommendations
- Improve financial awareness regarding recurring expenses

5. Proposed Solution

Users upload expense records (CSV/Excel) or enter them manually. Python + Pandas process the data to identify recurring subscriptions. Results are displayed via interactive Plotly visualizations, while the Groq API generates personalized recommendations.

The solution emphasizes simplicity, usability, and focused financial analysis.

6. System Architecture Overview



7. Technology Stack

Layer	Technology
Frontend	Streamlit
Programming Language	Python
Data Processing	Pandas
Visualization	Plotly
Spreadsheet Support	OpenPyXL
AI Recommendation Engine	Groq API
Development Environment	Visual Studio Code

8. Major Functional Features

- CSV/Excel file upload
- Manual expense entry

- Expense data processing
- Recurring subscription detection
- Subscription listing
- Monthly spending analysis
- Interactive dashboard
- Plotly visualizations
- Renewal reminders
- AI-powered recommendations via Groq API

9. Development Methodology

The project followed a structured lifecycle:

1. Idea selection and problem definition
2. Requirement analysis
3. Solution design
4. Implementation
5. Testing and validation
6. Documentation

10. Testing Summary

Testing verified:

- File upload and manual entry
- Data processing accuracy
- Subscription detection correctness
- Dashboard rendering
- Plotly visualization responsiveness
- AI recommendation generation
- Overall application stability

11. AI Integration Summary

- **Development Assistance:** ChatGPT supported planning, coding, debugging, and documentation.
- **Application Feature:** Groq API generates subscription optimization recommendations.

No custom AI model was trained; Groq API inference was used.

12. Project Deliverables

- Functional Streamlit application
- Subscription detection module
- Interactive dashboard
- Plotly visualizations
- AI recommendation module

- Technical documentation

13. Current Project Scope

Included: Expense upload, manual entry, subscription detection, dashboard, renewal reminders, AI recommendations. **Excluded:** Authentication, database integration, cloud deployment, bank API integration, email notifications, mobile app, multi-user support.

14. Project Outcomes

The project demonstrates practical AI application in BFSI by transforming raw expense data into actionable insights. Users can:

- Understand recurring subscription spending
- Detect unnecessary subscriptions
- Monitor upcoming renewals
- Receive AI-assisted recommendations
- Improve financial awareness

15. Future Enhancement Opportunities

Potential improvements:

- Authentication
- Database integration
- Cloud deployment
- Bank API integration
- Email notifications
- Advanced analytics
- Multi-user support
- Multi-currency support

16. Conclusion

The **Subscription Waste Detector** provides a focused, maintainable solution to recurring subscription expense management. By combining expense processing, subscription detection, visualization, reminders, and AI-powered recommendations, the application delivers meaningful financial insights.